

PRODUCT DOCUMENTATION

Clean-Label & E-Number Replacement Finder

User guide & technical documentation — a curated knowledge base of 36 food additives and 79 natural replacement swaps for clean-label reformulation, with label declarations, indicative trial dosages, functional-match grades, trade-offs and regulatory notes for every swap — plus three honest "keep as-is" verdicts where removal would be a mistake.

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Private by design: everything runs on the user's device — no uploads, no accounts, no tracking. Honest by design: dosages are labelled everywhere as starting points for trials, never guarantees; where a swap changes only the declaration and not the chemistry, the app says so in plain words; and where the right answer is to keep the additive, the app says that too.

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1. Overview — the clean-label reformulation problem

"Clean label" is not a legal category anywhere in the world. It is a market expectation: ingredient lists that shoppers recognise, names they can pronounce, and — increasingly — retailer and brand standards that name specific additives as unwanted. Meeting that expectation is a reformulation problem, and it is harder than it looks, because every additive on a label is there doing a job. Remove the antioxidant and the frying oil oxidises; remove the mould inhibitor and the bread's validated shelf life collapses; remove the emulsifier and the loaf loses volume. A swap that wins the label and loses the product is not a solution.

The practical difficulty is that the knowledge needed is scattered and asymmetrical. For each synthetic additive there may be two or three plausible natural alternatives, each with a different strength: one matches the function but flavours the product, one is flavour-neutral but weaker, one works only below a certain pH, one is authorised in the US but has an unsettled novel-food status in the EU. Dosage folklore travels by word of mouth; the difference between a declaration swap and a chemistry swap (Chapter 5, the nitrite and MSG examples) is routinely blurred in marketing material; and blanket "remove all E-numbers" briefs regularly target additives — citric acid, ascorbic acid, xanthan gum — that shoppers actually accept and that cost real product quality to remove.

This app gathers that knowledge into one searchable, curated database:

- **36 additives**, each profiled with its E-number, function(s), typical applications, origin, the factual reason brands move away from it, and a status note (legal maxima, warning labels, market bans where they exist).

- **79 replacement swaps**, each graded Strong or Partial, with the exact label declaration, an indicative dosage range, best-fit product categories, a one-line regulatory note, and the trade-off stated openly.
- **3 "keep as-is" verdicts** — citric acid (E330), ascorbic acid (E300) and xanthan gum (E415) — where the honest professional answer is do not reformulate (Chapter 6).
- **A reformulation-plan builder** that collects chosen swaps, with their declarations and dosages, for copying or CSV export — the working document of a reformulation project.

> **Guidance, not clearance.** This is a curated knowledge base built from established food-science and regulatory sources (Chapter 3). It is not legal advice, it does not replace supplier documentation, and no swap in it is validated until it has passed challenge tests and shelf-life trials in the actual product (Chapter 8). Regulatory notes are deliberately kept at EU/US level with "confirm locally" hedging.

2. How to use the finder

2.1 Search

The search box matches in every direction a formulator actually works:

- **By E-number** — with or without the "E": `319`, `e319` and `E319` all find TBHQ. The matcher normalises spaces, hyphens and the leading "E".
- **By additive name** — "sorbate", "nitrite", "carrageenan".
- **By replacement** — searching "rosemary" or "stevia" finds the additives those ingredients can replace, because replacement names and declarations are indexed too.
- **By function** — "antioxidant", "preservative".

A nonsense query returns a friendly "no matches" state with the counts intact, never a blank page.

2.2 Function filters

Ten filter chips — All, Preservative, Antioxidant, Emulsifier, Stabiliser & thickener, Colour, Sweetener, Acidity regulator, Raising agent, Flavour enhancer, Anti-caking & humectant — narrow the list to one functional family. Filters combine with search and with the category selector. Every filter returns at least one additive in the shipped database (verified in testing).

2.3 Product-category fit

The category selector (Bakery / Meat & seafood / Beverages / Sauces & dressings / Dairy & alternatives / Snacks / Confectionery) re-judges every replacement card against the chosen

context: each card is flagged **"Good fit"** or **"Poor fit"** from its curated best-fit list. A swap that is excellent in one matrix can be useless in another — celery powder is a meat fit and a beverage nonsense — and the flag makes that visible before the swap is added to a plan.

2.4 The detail panel

Selecting an additive opens its profile: functions, typical applications, origin, why brands move away, and the status note — followed by its replacement cards (Chapter 4), or, for the three keep-verdict entries, a green verdict box explaining why no swap is offered.

2.5 The reformulation plan

Each replacement card has an **Add to plan** button. The plan is a table — replaces → replacement → label declaration → dosage → match — with per-row removal, duplicate protection (the same swap cannot be added twice), a **Copy plan** action (plain text, ending with the validation disclaimer), a **Download plan CSV** action, and Clear. The plan lives only in the page: nothing is stored, nothing leaves the device, and reloading starts a fresh plan.

3. The database — 36 additives by function group

E-number ↔ name ↔ function assignments follow the EU Union list of food additives (Regulation (EC) No 1333/2008, Annex II) and standard references (Codex GSFA). Status anchors are matters of public record: E171 titanium dioxide was removed from EU authorisation in 2022 after EFSA's 2021 opinion (it remains permitted in the US); the "Southampton six" colours carry the mandatory EU warning "may have an adverse effect on activity and attention in children"; sulphites require allergen declaration above 10 mg/kg; aspartame's 2023 IARC Group 2B classification is recorded as a debate alongside JECFA's retained ADI, not as a ban. Dosages are conservative, commonly published starting ranges (Chapter 8).

E-no.	Additive	Function(s)	Swaps
E310	Propyl gallate	Antioxidant	2
E319	TBHQ (tert-butylhydroquinone)	Antioxidant	3
E320	BHA (butylated hydroxyanisole)	Antioxidant	2
E321	BHT (butylated hydroxytoluene)	Antioxidant	2
E200	Sorbic acid	Preservative	2
E202	Potassium sorbate	Preservative	3
E211	Sodium benzoate	Preservative	2
E220	Sulphur dioxide & sulphites (incl. E223)	Preservative, Antioxidant	2
E250	Sodium nitrite	Preservative, Colour	3
E251	Sodium nitrate	Preservative, Colour	2
E471	Mono- and diglycerides of fatty acids	Emulsifier	2
E472e	DATEM	Emulsifier	3
E481	SSL (sodium stearoyl-2-lactylate)	Emulsifier	3
E433	Polysorbate 80	Emulsifier	2
E405	Propylene glycol alginate (PGA)	Stabiliser & thickener, Emulsifier	2
E407	Carrageenan	Stabiliser & thickener	3
E415	Xanthan gum	Stabiliser & thickener, Emulsifier	Keep verdict
E466	CMC (carboxymethyl cellulose)	Stabiliser & thickener, Emulsifier	3
E1422	Modified starch (acetylated distarch adipate, for the E1404–E1452 family)	Stabiliser & thickener	3
E102	Tartrazine	Colour	2
E110	Sunset yellow FCF	Colour	2
E129	Allura red AC	Colour	3
E133	Brilliant blue FCF	Colour	2
E150c	Ammonia caramel (class III)	Colour	2
E171	Titanium dioxide	Colour	3
E950	Acesulfame K	Sweetener	2

E951	Aspartame	Sweetener	3
E954	Saccharin	Sweetener	2
E955	Sucralose	Sweetener	2
E300	Ascorbic acid (vitamin C)	Antioxidant, Acidity regulator	Keep verdict
E330	Citric acid	Acidity regulator, Antioxidant	Keep verdict
E338	Phosphoric acid	Acidity regulator	2
E450	Phosphate salts (diphosphates & triphosphates, E450/E451)	Acidity regulator, Raising agent, Stabiliser & thickener	2
E621	MSG (monosodium glutamate)	Flavour enhancer	3
E420	Sorbitol	Anti-caking & humectant, Sweetener	3
E551	Silicon dioxide (silica)	Anti-caking & humectant	2

36 additives · 79 swaps · 3 keep verdicts. The header counts in the app are computed from the database itself, not typed in — a database test asserts them (Chapter 9).

4. Reading a swap card

Every replacement is presented as a card with the same six fields, in the same order, so cards can be compared at a glance:

Field	What it tells you
Replacement name	The ingredient or system that takes over the function — sometimes a single ingredient, sometimes a system (e.g. "hurdle approach: pasteurisation + acidity + cold chain"), because some functions honestly have no single-ingredient answer.
Match grade	**Strong** — the replacement can carry the function at comparable performance in the stated categories. **Partial** — it covers part of the function, or covers it with a measurable penalty (shorter shelf life, narrower spectrum, texture change). Most preservative swaps are Partial, and the grade says so rather than overselling them.
Label declaration	How the replacement appears in the ingredient list — the actual commercial point of the exercise. Where the replacement itself carries an E-number (tocopherols E306, lecithin E322, stevia E960, rosemary extract E392), the declaration shows it: the gain is a recognisable name, not the disappearance of the numbering system (Chapter 7).
Indicative dosage	A conservative, commonly published starting range, always labelled "indicative starting point — validate in trials". Where dosing is supplier-standardised (celery powder to a nitrite-equivalent target, enzymes at ppm level), the card says "supplier-guided" instead of inventing a number.
Regulatory note	One hedged line on the EU/US position: authorised, GRAS, ingredient vs additive status, processing-aid caveats, novel-food flags where they exist. Guidance only — confirm category limits and local status before committing (Chapter 8).
Trade-off	The cost of the swap, stated plainly: flavour carry-over, colour contribution, efficacy ceiling, shorter validated shelf life, batch variability, or price. A card without a trade-off would be marketing, not documentation — every card has one.

> **Reading the profile above the cards.** The additive's own profile matters as much as the swaps: why brands move away separates evidence from fashion (a Prop 65 listing is a labelling trigger, not a food ban; a retailer avoid-list is a commercial fact, not a safety finding), and the status note records hard constraints — legal maxima, mandatory warnings, allergen thresholds — that any replacement must still satisfy.

5. Worked examples — five swaps in full

5.1 TBHQ (E319) → rosemary extract, mixed tocopherols, green tea extract

The additive. TBHQ is a synthetic antioxidant for frying oils, instant noodles, crackers, snack seasonings and edible fats, permitted in the EU and US with category limits (typically

up to 200 mg/kg on the fat) and not permitted in Japan. Brands remove it because the name reads as industrial chemistry and several retailers list it as an ingredient to avoid — a commercial driver, stated as such.

The lead swap — rosemary extract (Strong). Declared as "antioxidant: rosemary extract" (E392 where the E-number system is used). Indicative dosage **0.02–0.1% on the fat/oil basis (200–1,000 mg/kg of oil)**, standardised on carnosic acid content. EU: authorised as E392 with category limits; US: used as a natural antioxidant extract. Trade-off: a mild herbal note at high doses, higher cost, and supplier-to-supplier variability — standardise the specification, not just the name.

The supporting swaps. Mixed tocopherols ("antioxidant: mixed tocopherols", E306; 0.02–0.2% on the fat basis, start at 0.05%) are Strong in snacks, bakery and dressings but weaker than TBHQ in severe frying duty — they work best with citric acid as a metal chelator and disciplined oil management. Green tea extract (0.01–0.05%) is Partial: bitterness, astringency and colour cap the dose, and it is less proven in frying oils. The lesson the card teaches: the swap is a system (extract + chelator + oil handling), verified by oxidation testing, not a 1:1 ingredient exchange.

5.2 Sodium nitrite (E250) → celery powder — the swap that changes the declaration, not the chemistry

The additive. Nitrite fixes the cured colour of ham, bacon, frankfurters and salami and — the point that must never be lost in a labelling discussion — provides genuine protection against Clostridium botulinum. Strict legal maxima and residual limits apply in the EU and US. Removing nitrite is a food-safety decision, not only a label decision.

The swap — cultured celery powder (Strong, for the US route). Celery is naturally rich in nitrate, which cultures convert to nitrite; dosed at **0.2–0.5%, standardised to give an ingoing nitrite equivalent to local limits**, it cures. The database states the nuance in full, both ways:

> **The honest nuance.** Chemically, celery-derived nitrite is nitrite — the swap changes the declaration, not the chemistry. The markets disagree about the declaration: in the **US** (USDA), the route is accepted with the product labelled "Uncured" / "No nitrates or nitrites added except those naturally occurring". In the **EU**, the Commission/EFSA position is that vegetable-derived nitrite used for curing performs an additive function and is controlled as such — the celery route is generally not accepted as a way around E250. Any plan that exports must resolve this before artwork, not after.

The partners. Cherry/acerola powder (0.2–0.4%) is not a nitrite source — it is a cure accelerator, a natural ascorbate source that speeds colour formation, and is normally paired with celery powder; the card grades it Partial and says exactly that. Swiss chard juice powder is a second nitrate-rich vegetable route with the same nuance and less standardisation. Trade-offs across the entry: vegetable nitrite content varies by batch, so dose control is harder than with weighed E250, and colour and flavour differ slightly.

5.3 Modified starch (E1422, for the E1404–E1452 family) → pulse flours, native starches, citrus fibre

The additive. Acetylated distarch adipate stands in the app for the chemically modified starch family: freeze–thaw- and shear-stable thickening for sauces, soups, pie fillings, yoghurts and ready meals. "Modified" is one of the words consumers distrust most — even though the base is ordinary starch, and even though the freeze–thaw and retort performance is precisely why these starches persist in industrial recipes.

The swaps — all Partial, honestly. Functional pulse flours (chickpea / faba bean; "chickpea flour" as an ordinary ingredient declaration; **2–6% of recipe, replacing starch 1:1 by trial**) bring body plus a protein/fibre story, with slight flavour and colour and no suitability for clear sauces. Native starches (tapioca / potato / waxy maize; 2–5%) declare as plain "tapioca starch" with no E-number, but give thinner freeze–thaw stability and more syneresis — expect a shorter texture life. Citrus fibre (0.5–2%, often blended with native starch) adds water binding and a fibre story, with a slightly pulpy read in smooth sauces. The entry's teaching point: modified starch is a performance ingredient, and the clean-label cost is paid in texture stability — the plan should budget freeze–thaw trials, not just a label change.

5.4 Aspartame (E951) → stevia, monk fruit, erythritol blends

The additive. A synthetic high-intensity sweetener for diet drinks, sugar-free gum, yoghurts and desserts. In 2023 IARC classified aspartame "possibly carcinogenic" (Group 2B) while JECFA retained the ADI — the app records the headlines, not a ban, as the reformulation driver, and notes the EU's mandatory "contains a source of phenylalanine" statement. Aspartame is also not heat-stable, which is why it is absent from baked goods.

The swaps. Steviol glycosides ("steviol glycosides", E960 — itself an E-number; **0.01–0.05%** for high-purity grades; Partial) reverse the heat limitation: stevia survives baking, but the taste profile differs from aspartame and flavourings must be re-profiled. Monk fruit extract (0.01–0.05%; US GRAS) carries an explicit flag: **check novel-food status in the EU** — its European position has been unsettled, and the card refuses to assert it. Erythritol blends restore the bulk that high-intensity sweeteners cannot (erythritol 0.5–5% in dry mixes and desserts, with stevia or monk fruit), at ~70% of sugar's sweetness, with a cooling effect and a possible EU polyol-laxation labelling duty at high levels. The same unsettled-status honesty applies to allulose where it appears in the sweetener entries: flagged as not EU-authorised, never presented as a ready swap.

5.5 MSG (E621) → yeast extract — the declaration swap, named

The additive. Monosodium glutamate is fermentation-produced — yet perceived as synthetic. "No MSG" is one of the oldest clean-label claims, rooted in outdated anecdote rather than current safety assessments; the EU authorises E621 (general limit 10 g/kg) and the US treats it as GRAS with declaration. The app's status note frames the whole entry honestly: the swaps keep glutamate — from foods — while changing the declaration.

The lead swap — yeast extract (Strong). Declared simply as "yeast extract", a food ingredient in the EU/US rather than an additive, at **0.2–1% of recipe**. Its trade-off line is the most quoted sentence in the database: "Contains natural glutamates — that is precisely why it works. The swap changes the declaration, not the glutamate." It adds yeasty depth and is wrong for delicate products. The Partial partners work the same way: tomato powder (0.5–3%; glutamate-rich, but brings colour and acidity — tomato-compatible products only) and mushroom powder (0.3–1.5%; guanylate-rich umami synergy, earthy flavour and cost limiting the dose). A formulator who needs the glutamate gone — rather than renamed — will not find that here, and the app does not pretend otherwise.

6. When NOT to swap — the keep verdicts

Three of the 36 entries carry no swaps at all. Instead, the detail panel shows a green verdict box. This is a deliberate design position: a finder that only ever recommends change would push formulators into worse products, and a blanket "remove all E-numbers" brief is bad formulation policy.

Additive	Verdict (as shown in the app)
E330 — Citric acid	"Keep as-is. Citric acid is fermentation-produced, naturally present in citrus fruit, and recognised by shoppers when declared by name. Removing it gains nothing on the label ("citric acid" IS the friendly declaration) and loses pH control and antioxidant synergy. If a brief demands action, the only honest alternative is lemon juice concentrate — weaker, variable and flavour-active."
E300 — Ascorbic acid (vitamin C)	"Keep as-is. Ascorbic acid is vitamin C: shoppers read it as a nutrient, regulators treat it as one of the lowest-concern additives, and it actively protects colour and flavour. A "no E300" brief should be challenged — the swap would be to weaker, costlier antioxidant systems for zero label gain."
E415 — Xanthan gum	"Keep as-is. Xanthan is fermentation-derived, effective at very low doses (0.05–0.5%), and accepted by nearly all clean-label standards. Removing it usually costs more texture quality than the label gains. If a retailer standard excludes it, guar gum or citrus fibre can cover some uses — at weaker suspension performance."

Note what the verdicts share: each names the cost of removal (lost pH control, lost protection, lost texture) against a label gain of zero, and each still tells the user what the fallback is if an external standard forces the issue. Challenging a brief is part of the tool's job.

7. Label-declaration principles

- **An E-number is an authorisation code, not a hazard grade.** It means the substance was safety-assessed and authorised with conditions. Vitamin C is E300;

lecithin is E322; stevia extract is E960. The number tells a shopper nothing about risk — which is exactly why declaration names, not numbers, drive clean-label work.

- **Natural replacements may carry E-numbers — and the app shows them.** Rosemary extract is E392, mixed tocopherols E306, sunflower lecithin E322, steviol glycosides E960, erythritol E968. A swap card that hid the number would be teaching the user a falsehood; the gain being bought is a familiar declaration ("antioxidant: rosemary extract"), and the card prices it honestly.
- **"Clean label" is not a legal category.** No regulator approves the term. What exists are (a) additive law — what may be used, where, and at what level; (b) labelling law — how it must be declared, including allergen and warning duties that survive any swap (sulphites above 10 mg/kg, the Southampton colours warning, polyol laxation statements, the phenylalanine statement); and (c) private standards — retailer and brand lists, which are commercial facts. The database keeps the three separate.
- **Declaration swaps and chemistry swaps are different products.** Celery-for-nitrite and yeast-extract-for-MSG change the words on the pack while the functional chemistry (nitrite, glutamate) remains. That can be a legitimate labelling strategy where the market accepts it — the US "uncured" route — and an unacceptable one where it does not — the EU position on vegetable nitrite. What it must never be is unlabelled: if a consumer avoiding nitrite or glutamate would be misled, the strategy fails ethically before it fails legally. The database flags both cases in the swap text itself.
- **Processing aids and ingredients are different declaration worlds.** Bakery enzyme systems often need no ingredient declaration (status differs by market — the cards say "confirm"); cultured fermentates declare as ingredients; colouring foods declare as foods. Two swaps with identical function can have opposite label outcomes, which is why every card leads with its declaration.

8. Honest scope and limitations

- **Dosages are starting points, never specifications.** Every dosage in the database is a conservative, commonly published range for first trials, labelled as such in the app. Supplier grades vary (rosemary standardised on carnosic acid; celery powder standardised on nitrite equivalence), matrices vary, and the correct dose is the one established by the user's own trials.
- **Efficacy must be proven, not presumed.** Preservative swaps in particular are graded Partial because fermentates and acids are narrower-spectrum than sorbate/benzoate systems. A reformulated product needs challenge testing (including preservative-resistant yeasts where relevant) and validated shelf-life studies — the plan export ends with this disclaimer, deliberately.
- **Regulatory status changes — the E171 example.** Titanium dioxide was a routine whitener until EFSA's 2021 opinion and its removal from EU authorisation in 2022; it remains permitted in the US. Any database of "what is allowed" is a snapshot. The

app names its anchors and hedges its notes ("generally authorised; confirm category limits"); users must confirm current status in each destination market, and unsettled statuses (monk fruit and allulose in the EU) are flagged in the cards rather than resolved by assertion.

- **One-line regulatory notes are guidance, not legal advice.** Category maxima, residual limits and declaration duties are product- and market-specific. Regional systems beyond the EU/US (including GSO) are deliberately not asserted.
- **Concern is recorded as fact, not amplified.** Where a removal driver is reputational (retailer lists, Prop 65 labelling triggers, headline classifications), the profile says so, and distinguishes it from safety findings (the carrageenan note separates food-grade carrageenan from poligeenan, a different, non-food substance).
- **Coverage is curated, not exhaustive.** 36 additives and 79 swaps cover the common clean-label briefs; absence of an additive or a swap is not a judgement that none exists.

9. How correctness was verified — testing summary

The app's PURE CORE (database + search/filter/plan logic, Appendix) was extracted from the shipped page and run in Node against hand-set expectations; both inline scripts also passed `node --check`. **69 / 69 checks passed.**

Group	Checks	What was asserted
Data integrity	10	Database size is exactly 36; E-numbers unique and format-valid (3–4 digits — the modified-starch family is genuinely 4-digit, e.g. E1422); every additive fully profiled; every additive has ≥1 replacement or a keep verdict; every replacement fully specified (declaration, dosage, Strong/Partial match, valid categories, regulatory note, trade-off).
Factual anchors	24	E319 TBHQ → rosemary with the "0.02…0.1" dosage string, tocopherols, green tea; E250 nitrite → celery with both US and EU positions and the "chemically the same nitrite" nuance, all nitrite swaps meat-fit; E171 status cites EU + 2022; E621 MSG with the honest glutamate note and yeast extract; E960 stevia replaces ≥2 sweeteners; E322 lecithin appears as a replacement; modified starch → pulse flours; DATEM → lecithin + enzymes; sorbitol → erythritol with the laxation note; silica → rice anti-caking; E129 → colouring foods; the Southampton warning on tartrazine.
Keep verdicts	4	Exactly 3 keep verdicts — E330, E300, and E415 with its fermentation note.
Search & filters	17	"319" / "e319" / "E319" all find TBHQ; replacement-name search works (rosemary, stevia; "sorbat" → E202); nonsense query returns empty; empty query returns all 36; TBHQ sits under Antioxidant and NOT under Colour; every function filter returns ≥1 additive; category fit flags behave (celery fits Meat & seafood, poor fit for Beverages; colouring foods fit Beverages and Confectionery somewhere in the database).
Plan builder	14	Three adds give three unique rows; a duplicate add is rejected; unknown additive/replacement pairs are rejected; the CSV is a header + 3 rows containing the declarations; the plan text lists all swaps and ends with the challenge-test disclaimer; remove deletes exactly one row; empty-plan text and CSV are friendly, not errors.

The full test report (`clean-label-replacement-finder-test-report.md`) ships with the app in the GitHub repository, including the entries where facts needed care and how each was hedged.

10. Adding the product to the Asif SaaS Studio catalogue (Wix CMS)

1. **Open the item.** In the Wix dashboard, open the CMS collection **SaasProducts** and edit (or add) the item with slug `clean-label-replacement-finder`.
2. **Paste the app.** Paste the full contents of `index.html` into the **appHtml** field; set Title, Category (Food Science), Version 1.0, Published = on.
3. **Save and open the live page**, then run the smoke test below.

Listing field	Value
Title	Clean-Label & E-Number Replacement Finder
Description	Search 36 additives and 79 natural swaps — with label declarations, indicative dosages, trade-offs and regulatory notes for clean-label reformulation.
Category	Food Science
Slug	`clean-label-replacement-finder`
Version	1.0

Smoke test (expected values). The header counts read **"36 additives · 79 swaps · 3 honest "keep as-is" verdicts"** and the detail panel opens on **E319 TBHQ** — its first replacement card is **Rosemary extract, Strong match, dosage 0.02–0.1% on the fat/oil basis**. Search `250` → **E250 Sodium nitrite** appears with the celery powder swap and the US/EU nuance. Pick function **Colour** → TBHQ disappears and E171 Titanium dioxide is listed. Add the rosemary swap to the plan → the plan table shows 1 row whose declaration mentions rosemary extract (E392).

11. Product information and credits

Item	Detail
Product	Clean-Label & E-Number Replacement Finder, version 1.0
Creator	Dr. Asif Ahmad — Asif SaaS Studio
Format	One self-contained `index.html` (62,632 bytes): all CSS and JavaScript inline; no build step, no server, no external assets; no localStorage, service workers, browser permissions or parent-window access; runs fully offline and inside a sandboxed iframe.
Knowledge basis	EU Union list of additives (Regulation (EC) No 1333/2008), Codex GSFA, EFSA opinions and EU authorisation changes of public record, USDA/US labelling practice for natural curing, and commonly published supplier starting ranges for the replacement ingredients — curated and hedged as described in Chapters 3 and 8.
Testing	69 automated checks against the shipped core — 69 passed (Chapter 9).
Licence	MIT (repository).

Created by Dr. Asif Ahmad · Asif SaaS Studio · v1.0 · October 2026

12. Appendix — the tested core (data + logic), verbatim

The complete PURE CORE of the shipped app — the curated database (36 additives, 79 swaps, 3 keep verdicts) and the search / filter / plan logic — exactly as tested. If the documentation and this appendix ever disagree, the appendix is the product.

```
/*PURE-CORE-START*/
/* ============ Clean-Label Replacement Finder — PURE CORE (no DOM) ============
   Curated knowledge base + search / filter / plan logic.
   Data conventions:
   Additive: {e, n, f[], ap, o, w, s, keep, r[]}
     e = E-number ("—" if none), n = name, f = functions, ap = typical applications,
     o = origin, w = why brands move away, s = status note, keep = "keep as-is" verdict text or null,
     r = replacements.
   Replacement: {n, d, dose, m, c[], g, t}
     n = name, d = label declaration, dose = indicative starting range,
     m = "Strong" or "Partial", c = best-fit categories, g = regulatory note, t = trade-off.
============================================================================== */
var FUNCTIONS = ["Preservative","Antioxidant","Emulsifier","Stabiliser &
thickener","Colour","Sweetener","Acidity regulator","Raising agent","Flavour enhancer","Anti-caking &
humectant"];
var CATEGORIES = ["Bakery","Meat & seafood","Beverages","Sauces & dressings","Dairy &
alternatives","Snacks","Confectionery"];
var DB = [
/* ---- Antioxidants ---- */
{e:"E319",n:"TBHQ (tert-butylhydroquinone)",f:["Antioxidant"],ap:"Frying oils, instant noodles, crackers, snack
seasonings, edible fats",o:"Synthetic",w:"A synthetic-sounding name consumers increasingly reject; several
retailers list it as an ingredient to avoid.",s:"Permitted in the EU and US with category limits (typically up
to 200 mg/kg on the fat). Not permitted in Japan.",keep:null,r:[
{n:"Rosemary extract",d:"“Antioxidant: rosemary extract” (may be declared as E392 where the E-number system is
used)",dose:"0.02–0.1% on the fat/oil basis (200–1,000 mg/kg of oil)",m:"Strong",c:["Snacks","Meat &
seafood","Sauces & dressings"],g:"EU: authorised as E392 with category limits. US: used as a natural flavouring/
antioxidant extract.",t:"Mild herbal note at high doses; cost per kg is higher and quality varies by supplier —
standardise on carnosic acid content."},
{n:"Mixed tocopherols",d:"“Antioxidant: mixed tocopherols” (E306)",dose:"0.02–0.2% on the fat basis — start at
0.05%",m:"Strong",c:["Snacks","Bakery","Sauces & dressings"],g:"EU: E306, quantum satis in most foods. US: GRAS.
Still an E-number — the gain is a recognisable declaration.",t:"Weaker than TBHQ in severe frying duty; works
best combined with citric acid (metal chelation) and good oil management."},
{n:"Green tea extract",d:"“Green tea extract” (no E-number as a food ingredient in most markets)",dose:"0.01–
0.05% on the fat basis",m:"Partial",c:["Snacks","Beverages"],g:"Generally accepted as a food extract in the EU/
US; catechin content and any claims must be checked locally.",t:"Can add bitterness/astringency and colour at
higher doses; less proven than rosemary in frying oils."}]},
{e:"E320",n:"BHA (butylated hydroxyanisole)",f:["Antioxidant"],ap:"Cereals, chewing gum, dehydrated potatoes,
fats and oils, snack foods",o:"Synthetic",w:"Long-running consumer concern and retailer avoid-lists; California
lists BHA as a carcinogen under Proposition 65 (a labelling trigger, not a food ban).",s:"Permitted in the EU
(E320) and US with limits; carries a Prop 65 warning obligation in California.",keep:null,r:[
{n:"Mixed tocopherols",d:"“Antioxidant: mixed tocopherols” (E306)",dose:"0.02–0.2% on the fat
basis",m:"Strong",c:["Snacks","Bakery","Confectionery"],g:"EU: E306. US: GRAS.",t:"Slightly less potent per unit
weight; pair with rosemary for synergy."},
{n:"Rosemary extract",d:"“Antioxidant: rosemary extract” (E392)",dose:"0.02–0.1% on the fat basis",m:"Strong",c:
["Snacks","Meat & seafood"],g:"EU: E392 with category limits. US: accepted natural extract.",t:"Herbal carry-
over possible in delicate products; deodorised grades cost more."}]},
{e:"E321",n:"BHT (butylated hydroxytoluene)",f:["Antioxidant"],ap:"Packaging-adjacent antioxidant for cereals,
snacks, chewing gum, oils",o:"Synthetic",w:"Frequently named on consumer avoid-lists; some markets and retailers
restrict it.",s:"Permitted in the EU (E321) and US with limits; consumer pressure is the main driver for
removal.",keep:null,r:[
{n:"Mixed tocopherols",d:"“Antioxidant: mixed tocopherols” (E306)",dose:"0.02–0.2% on the fat
basis",m:"Strong",c:["Snacks","Bakery","Confectionery"],g:"EU: E306. US: GRAS.",t:"Protects packaged foods less
aggressively under heat stress — verify with accelerated shelf-life tests."},
{n:"Rosemary extract",d:"“Antioxidant: rosemary extract” (E392)",dose:"0.02–0.1% on the fat basis",m:"Strong",c:
["Snacks","Sauces & dressings"],g:"EU: E392. US: accepted natural extract.",t:"Flavour threshold is the ceiling
in bland products."}]},
{e:"E310",n:"Propyl gallate",f:["Antioxidant"],ap:"Fats and oils, soups, chewing gum, meat
products",o:"Synthetic",w:"Synthetic gallate ester with low consumer recognition; often removed together with
BHA/BHT.",s:"Permitted in the EU (E310) and US with limits, usually in blends with BHA/BHT and citric
acid.",keep:null,r:[
{n:"Rosemary extract",d:"“Antioxidant: rosemary extract” (E392)",dose:"0.02–0.1% on the fat basis",m:"Strong",c:
["Meat & seafood","Snacks"],g:"EU: E392. US: accepted natural extract.",t:"Check colour contribution in pale
fats."},
{n:"Mixed tocopherols + citric acid",d:"“Antioxidant: mixed tocopherols” (E306); citric acid (E330) as
chelator",dose:"Tocopherols 0.05–0.1% on fat; citric acid 0.005–0.02%",m:"Partial",c:["Sauces &
dressings","Snacks"],g:"Both widely authorised (E306/E330).",t:"The blend replaces the synergy gallates
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provided; match performance only after oxidation testing (e.g. OSI/Rancimat comparison)."}]},
/* ---- Preservatives ---- */
{e:"E200",n:"Sorbic acid",f:["Preservative"],ap:"Cheese, bakery fillings, beverages, dried fruit,
wine",o:"Nature-identical (also occurs in rowan berries)",w:"An E-number many brands remove for label
simplicity, although sorbic acid is one of the better-studied, lower-concern preservatives.",s:"Permitted
worldwide with category limits; most effective below pH 6.5. Honest view: a low-priority removal — see the
cultured alternatives if the label must change.",keep:null,r:[
{n:"Cultured dextrose / cultured wheat flour",d:"“Cultured dextrose” or “cultured wheat flour” (declared as an
ingredient, no E-number)",dose:"0.5–2% of the recipe — supplier-guided",m:"Partial",c:["Bakery","Dairy &
alternatives","Sauces & dressings"],g:"Marketed as food ingredients in the EU/US; status and naming differ by
market — confirm with supplier documentation.",t:"Fermentate activity varies; weaker and narrower-spectrum than
sorbic acid. Proof with challenge tests is essential."},
{n:"Vinegar (buffered)",d:"“Vinegar” or “buffered vinegar”",dose:"0.1–1% depending on pH target",m:"Partial",c:
["Sauces & dressings","Bakery"],g:"A conventional food ingredient everywhere.",t:"Only works where acidity is
acceptable; flavour impact in delicate products."}]},
{e:"E202",n:"Potassium sorbate",f:["Preservative"],ap:"Bakery goods, soft drinks, cheese, margarine, fruit
preparations",o:"Synthetic salt of a nature-identical acid",w:"The most common “E-number to remove” in bakery
briefs; consumers read it as a chemical preservative.",s:"Permitted worldwide with category limits; anti-mould
workhorse in bakery. Any replacement is weaker — plan shelf-life accordingly.",keep:null,r:[
{n:"Cultured wheat flour",d:"“Cultured wheat flour” (ingredient declaration, no E-number)",dose:"0.5–2% of flour
weight — supplier-guided",m:"Partial",c:["Bakery"],g:"Sold as a clean-label ingredient in EU/US; confirm local
status and naming.",t:"Good against mould in bread but not a like-for-like swap: expect shorter validated shelf
life and run mould challenge tests."},
{n:"Raisin juice concentrate",d:"“Raisin juice concentrate”",dose:"1–3% of the recipe in sweet bakery/
fillings",m:"Partial",c:["Bakery","Confectionery"],g:"A conventional food ingredient.",t:"Naturally occurring
sorbic/organic acids give mild inhibition only; adds sugars and colour — suited to sweet goods, not lean
bread."},
{n:"Vinegar (buffered)",d:"“Vinegar” / “buffered vinegar”",dose:"0.1–0.5% in bread systems with pH
control",m:"Partial",c:["Bakery","Sauces & dressings"],g:"A conventional food ingredient
everywhere.",t:"Suppresses rope and mould modestly; overuse flavours the crumb."}]},
{e:"E211",n:"Sodium benzoate",f:["Preservative"],ap:"Carbonated and still soft drinks, juices, pickles, sauces,
jams",o:"Synthetic salt of benzoic acid (which occurs naturally in berries)",w:"Focus of consumer concern in
drinks (including the benzene-formation debate when combined with ascorbic acid under heat/light); many drink
brands reformulate it out.",s:"Permitted in EU/US with limits (commonly 150 mg/L in EU soft drinks). Effective
only in acidic products (pH < 4.5).",keep:null,r:[
{n:"Cultured sugar / fermentate systems",d:"“Cultured sugar” (ingredient declaration)",dose:"0.1–0.5% in
beverages — supplier-guided",m:"Partial",c:["Beverages","Sauces & dressings"],g:"Ingredient status varies by
market; obtain supplier regulatory statements.",t:"Narrower spectrum than benzoate/sorbate systems; validate
against yeasts (incl. Zygosaccharomyces) with challenge tests."},
{n:"Hurdle approach: pasteurisation + lemon/vinegar acidity + cold chain",d:"No preservative declaration —
process and pH do the work",dose:"Process-defined, not a dose",m:"Partial",c:["Beverages"],g:"No additive
involved; hygiene and process rules apply instead.",t:"Shorter chilled shelf life and higher process cost; not
suitable for ambient long-life drinks without validation."}]},
{e:"E220",n:"Sulphur dioxide & sulphites (incl. E223 sodium metabisulfite)",f:
["Preservative","Antioxidant"],ap:"Wine, dried fruit, fruit juices, crustaceans, sausages, dehydrated
vegetables",o:"Mineral / synthetic",w:"Sulphites are allergens for a sensitive minority and must be declared
above 10 mg/kg — many brands reduce or remove them for both label and allergen reasons.",s:"Allergen labelling
is mandatory above 10 mg/kg SO2 in the EU and US. Permitted levels are tightly category-specific.",keep:null,r:[
{n:"Ascorbic acid + citric acid (anti-browning system)",d:"“Ascorbic acid” (E300) and “citric acid” (E330) —
kitchen-recognisable names",dose:"0.05–0.3% ascorbic acid in dips/cut fruit; product-specific",m:"Partial",c:
["Beverages","Snacks"],g:"E300/E330 are among the most accepted additives in both EU and US.",t:"Controls
browning, not microbes: sulphites' antimicrobial role in wine and dried fruit has no simple natural equal —
expect shorter life and colour drift."},
{n:"Lemon juice concentrate",d:"“Lemon juice concentrate”",dose:"0.5–2% in fruit preparations",m:"Partial",c:
["Beverages","Confectionery"],g:"A conventional food ingredient.",t:"Mild antioxidant/acid effect only; flavour
and acidity shift."}]},
{e:"E250",n:"Sodium nitrite",f:["Preservative","Colour"],ap:"Cured meats — ham, bacon, frankfurters, salami
(colour fixing + anti-botulinal protection)",o:"Synthetic (mineral-derived)",w:"Nitrite is scrutinised over
nitrosamine formation; demand for “no added nitrite” meats is high. But nitrite also provides genuine
Clostridium botulinum protection — removal is a safety decision, not just a label decision.",s:"Strict legal
maxima and residual limits apply in cured meats (EU and US). The celery route below is nitrite from a vegetable
source: chemically it IS nitrite once converted, and regulators treat the two markets differently.",keep:null,r:
[
{n:"Celery powder / celery juice (cultured)",d:"US: “celery powder” with the product labelled “Uncured” / “No
nitrates or nitrites added except those naturally occurring”. EU: generally NOT accepted as a way to add nitrite
function — nitrite from any source is regulated as an additive use.",dose:"0.2–0.5% celery powder, standardised
to give ingoing nitrite equivalent to local limits — supplier-guided",m:"Strong",c:["Meat & seafood"],g:"US
(USDA): permitted route for “natural/uncured” claims with the qualifying statement. EU: EFSA/Commission position
is that vegetable nitrite used for curing is an additive function — expect E250-style control.",t:"The honest
nuance: this swaps the declaration, not the chemistry — nitrate in celery is converted to the same nitrite. Dose
control is harder because vegetable nitrite content varies; colour and flavour differ slightly."},
{n:"Cherry powder (acerola) — cure accelerator partner",d:"“Cherry powder” / “acerola cherry powder”",dose:"0.2–
0.4% alongside celery powder",m:"Partial",c:["Meat & seafood"],g:"A food ingredient; used as a natural ascorbate
source.",t:"Not a nitrite source itself — it accelerates curing colour (as ascorbate/erythorbate does) and is
normally paired with celery powder."},
{n:"Swiss chard juice powder",d:"“Swiss chard juice powder”",dose:"Supplier-guided to nitrite-equivalent
target",m:"Partial",c:["Meat & seafood"],g:"US: used in natural curing blends. EU: same additive-function caveat
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as celery.",t:"Another nitrate-rich vegetable route — same chemical nuance and variability as celery; less
widely standardised."}]},
{e:"E251",n:"Sodium nitrate",f:["Preservative","Colour"],ap:"Slow-cured and fermented meats, some cheeses
(reservoir that converts to nitrite)",o:"Synthetic (mineral-derived)",w:"Same nitrosamine debate as nitrite;
traditional slow-cure products are under clean-label pressure.",s:"Permitted with category maxima in the EU/US;
converts to nitrite during ripening.",keep:null,r:[
{n:"Celery powder (nitrate-rich)",d:"US: “celery powder”, with “uncured” qualifying statements. EU: additive-
function caveat as for E250.",dose:"0.2–0.5%, standardised",m:"Partial",c:["Meat & seafood"],g:"US route
accepted with qualifying labelling; EU treats functional nitrite addition as additive use.",t:"Conversion rate
to nitrite depends on starter cultures and time — process control becomes critical."},
{n:"Swiss chard / vegetable nitrate blends",d:"“Vegetable juice powder” blends",dose:"Supplier-
guided",m:"Partial",c:["Meat & seafood"],g:"As celery: US natural-cure route; EU caveat applies.",t:"Less
standardised than celery; batch variability is the main risk."}]},
/* ---- Emulsifiers ---- */
{e:"E472e",n:"DATEM (diacyltartaric acid esters of mono- and diglycerides)",f:["Emulsifier"],ap:"Bread and rolls
(dough strength, volume), cakes, coffee whiteners",o:"Synthetic ester",w:"A long chemical name with an E-number;
one of the first targets in bread clean-label briefs.",s:"Permitted in the EU/US with bakery limits. Functional
loss on removal is real — replacement usually needs a system, not a single ingredient.",keep:null,r:[
{n:"Sunflower lecithin",d:"“Sunflower lecithin” (E322 — still an E-number, but a familiar, allergen-friendly
declaration; soy lecithin must declare SOY as an allergen)",dose:"0.3–1% on flour weight",m:"Partial",c:
["Bakery","Confectionery"],g:"EU: E322 authorised. US: GRAS; sunflower grades avoid the soy allergen
declaration.",t:"Weaker dough-strengthening than DATEM — expect some volume loss unless combined with
enzymes."},
{n:"Enzyme systems (lipase / phospholipase, ± xylanase, glucose oxidase)",d:"Usually a processing aid — often no
label declaration; flour may be declared as “enzyme-treated flour” in some markets",dose:"ppm levels (typically
5–50 mg/kg flour) — supplier-guided",m:"Strong",c:["Bakery"],g:"Processing-aid status differs by market; confirm
declaration duties with your enzyme supplier.",t:"The professional route for industrial bread: strong functional
match, but requires bake trials and careful dosing."},
{n:"Pulse proteins (chickpea / faba bean)",d:"“Chickpea flour” or “faba bean protein”",dose:"1–3% on flour
weight",m:"Partial",c:["Bakery","Snacks"],g:"Conventional food ingredients.",t:"Contribute emulsification plus
protein marketing value; beany notes possible at higher levels."}]},
{e:"E481",n:"SSL (sodium stearoyl-2-lactylate)",f:["Emulsifier"],ap:"Bread (softness, volume), tortillas,
pancakes, coffee whiteners",o:"Synthetic ester",w:"Chemical-sounding name; frequently removed alongside DATEM in
bakery reformulation.",s:"Permitted in EU/US with limits.",keep:null,r:[
{n:"Sunflower lecithin",d:"“Sunflower lecithin” (E322)",dose:"0.3–1% on flour weight",m:"Partial",c:
["Bakery"],g:"EU: E322. US: GRAS; sunflower avoids soy allergen labelling.",t:"Softer crumb effect is weaker;
anti-staling benefit is smaller."},
{n:"Enzymes (amylase + lipase systems)",d:"Processing aid — often no declaration (market-dependent)",dose:"ppm
levels — supplier-guided",m:"Strong",c:["Bakery"],g:"Confirm processing-aid status per market.",t:"Maltogenic
amylases replace SSL's anti-staling role well; combined systems need trials."},
{n:"Chickpea protein / aquafaba solids",d:"“Chickpea protein”",dose:"1–2% on flour weight",m:"Partial",c:
["Bakery","Sauces & dressings"],g:"Conventional food ingredient.",t:"Adds protein story; flavour and colour need
masking in white bread."}]},
{e:"E471",n:"Mono- and diglycerides of fatty acids",f:["Emulsifier"],ap:"Bread, cakes, ice cream, margarine,
peanut butter (anti-staling, crumb softening)",o:"Made from fats/oils (plant or animal) by glycerolysis — “semi-
natural” perception problem is mostly the name",w:"Declared in full it reads as a chemical; consumer groups have
questioned it, though safety assessments are favourable.",s:"Among the most widely permitted emulsifiers (EU
E471; US GRAS). Clean-label pressure is moderate.",keep:null,r:[
{n:"Sunflower / rapeseed lecithin",d:"“Sunflower lecithin” (E322)",dose:"0.2–0.5% of recipe (ice cream); 0.3–1%
flour basis (bakery)",m:"Partial",c:["Bakery","Dairy & alternatives","Confectionery"],g:"EU: E322. US:
GRAS.",t:"Does not replicate the amylose-complexing anti-staling effect; softness retention is shorter."},
{n:"Enzyme anti-staling systems (maltogenic amylase)",d:"Processing aid — often no declaration",dose:"ppm levels
— supplier-guided",m:"Partial",c:["Bakery"],g:"Market-dependent processing-aid status.",t:"Covers the bread use-
case well; does nothing for ice cream or spreads."}]},
{e:"E433",n:"Polysorbate 80 (polyoxyethylene sorbitan monooleate)",f:["Emulsifier"],ap:"Ice cream, non-dairy
creamers, pickles, whipped toppings, some supplements",o:"Synthetic",w:"A synthetic-sounding name that appears
on many consumer avoid-lists and retailer restricted lists.",s:"Permitted in EU/US with category limits; notable
for label-driven removal rather than new safety findings.",keep:null,r:[
{n:"Quillaja extract",d:"“Quillaja extract” (E999 where the E-number system applies)",dose:"0.01–0.05% in
beverages/foams",m:"Partial",c:["Beverages","Dairy & alternatives"],g:"EU: E999 with limits (mainly drinks). US:
permitted flavour/foaming agent in beverages.",t:"Excellent natural foaming/emulsifying saponin, but bitter at
excess and mainly a drinks solution."},
{n:"Sunflower lecithin",d:"“Sunflower lecithin” (E322)",dose:"0.2–0.5% of recipe",m:"Partial",c:["Dairy &
alternatives","Confectionery"],g:"EU: E322. US: GRAS.",t:"In ice cream it changes meltdown and overrun behaviour
— expect rebalancing with stabilisers."}]},
/* ---- Stabilisers & thickeners ---- */
{e:"E1422",n:"Modified starch — acetylated distarch adipate (representative of the modified-starch family E1404–
E1452)",f:["Stabiliser & thickener"],ap:"Sauces, soups, pie fillings, yoghurts, ready meals (freeze–thaw and
shear-stable thickening)",o:"Chemically modified plant starch",w:"“Modified” is one of the words consumers
distrust most, even though the base is ordinary starch.",s:"Each modified starch has its own E-number and spec;
all are permitted with category rules in the EU/US. Performance under freeze–thaw and retort is why they
persist.",keep:null,r:[
{n:"Functional pulse flours (chickpea / faba bean)",d:"“Chickpea flour” / “faba bean flour” — declared as
ordinary ingredients",dose:"2–6% of recipe, replacing starch 1:1 by trial",m:"Partial",c:["Sauces &
dressings","Snacks","Meat & seafood"],g:"Conventional food ingredients everywhere.",t:"Bring body plus protein/
fibre; slight flavour and colour; not a drop-in for clear sauces."},
{n:"Native starches (tapioca / potato / waxy maize)",d:"“Tapioca starch” / “potato starch” / “corn
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starch”",dose:"2–5% of recipe",m:"Partial",c:["Sauces & dressings","Dairy &
alternatives","Confectionery"],g:"Conventional food ingredients (no E-number).",t:"Thinner freeze–thaw stability
and more syneresis; waxy types perform best. Expect a shorter texture life."},
{n:"Citrus fibre",d:"“Citrus fibre”",dose:"0.5–2% of recipe, often blended with native starch",m:"Partial",c:
["Sauces & dressings","Meat & seafood","Dairy & alternatives"],g:"A food ingredient (fibre) in EU/US; confirm
fibre-claim rules if you declare fibre content.",t:"Adds water binding and a “fibre” story; can read slightly
pulpy in smooth sauces."}]},
{e:"E466",n:"CMC (carboxymethyl cellulose / cellulose gum)",f:["Stabiliser & thickener","Emulsifier"],ap:"Ice
cream, beverages, bakery fillings, gluten-free baking, sauces",o:"Semi-synthetic (from plant
cellulose)",w:"“Cellulose gum” attracts clean-label briefs, especially in ice cream and drinks.",s:"Permitted in
EU/US; also serves as the gluten-free baker's body-builder, where removal is hardest.",keep:null,r:[
{n:"Citrus fibre",d:"“Citrus fibre”",dose:"0.5–1.5% of recipe",m:"Partial",c:["Dairy & alternatives","Sauces &
dressings","Bakery"],g:"Food ingredient in EU/US.",t:"Good water binding; less clean viscosity curve than CMC,
slight pulp note."},
{n:"Guar gum",d:"“Guar gum” (E412 — a natural gum that still has an E-number)",dose:"0.1–0.5% of
recipe",m:"Partial",c:["Dairy & alternatives","Bakery","Beverages"],g:"EU: E412. US: GRAS.",t:"A plant-seed gum
most consumers accept; can be slimy if overdosed in drinks."},
{n:"Chia or flax mucilage systems",d:"“Chia seed” / “ground flaxseed”",dose:"1–3% of recipe",m:"Partial",c:
["Bakery","Dairy & alternatives"],g:"Conventional food ingredients.",t:"Best in rustic/gluten-free products;
visible seeds unless milled fine."}]},
{e:"E407",n:"Carrageenan",f:["Stabiliser & thickener"],ap:"Chocolate milk, dairy and plant drinks, ice cream,
processed meats, desserts (gelling protein-reactive gum)",o:"Seaweed extract (natural origin, processed)",w:"The
most debated gum in clean-label work: social-media-driven concern is high despite regulatory re-evaluations
keeping it authorised; many dairy brands removed it pre-emptively.",s:"Authorised in the EU (E407) and US;
degraded carrageenan (poligeenan) is a different, non-food substance and not what is used in food. Removal is
usually a marketing decision.",keep:null,r:[
{n:"Gellan gum",d:"“Gellan gum” (E418)",dose:"0.02–0.15% in drinks; 0.1–0.3% in gels",m:"Partial",c:
["Beverages","Dairy & alternatives"],g:"EU: E418. US: GRAS for many uses.",t:"Excellent suspension in drinks at
very low dose; gel texture is more brittle than carrageenan's elastic gel."},
{n:"Pectin",d:"“Pectin” (E440)",dose:"0.2–0.8% depending on system",m:"Partial",c:["Dairy &
alternatives","Confectionery","Beverages"],g:"EU: E440, among the most consumer-accepted gums. US:
GRAS.",t:"Works beautifully in acidified dairy and fruit systems; not a direct chocolate-milk suspension swap on
its own."},
{n:"Locust bean gum (LBG) blends",d:"“Locust (carob) bean gum” (E410)",dose:"0.1–0.4%, usually blended with
xanthan or guar",m:"Partial",c:["Dairy & alternatives","Sauces & dressings"],g:"EU: E410. US: GRAS.",t:"Blend-
dependent; supply and price of LBG are volatile."}]},
{e:"E415",n:"Xanthan gum",f:["Stabiliser & thickener","Emulsifier"],ap:"Dressings, gluten-free baking, sauces,
beverages",o:"Fermentation-derived (bacterial fermentation of sugars)",w:"Occasionally questioned because of the
name — yet it is produced by fermentation, the same family of processes consumers accept in cultured
foods.",s:"Authorised in the EU (E415) and US (GRAS). Most clean-label standards accept xanthan. Typical advice:
keep.",keep:"Keep as-is. Xanthan is fermentation-derived, effective at very low doses (0.05–0.5%), and accepted
by nearly all clean-label standards. Removing it usually costs more texture quality than the label gains. If a
retailer standard excludes it, guar gum or citrus fibre can cover some uses — at weaker suspension
performance.",r:[]},
{e:"E405",n:"Propylene glycol alginate (PGA)",f:["Stabiliser & thickener","Emulsifier"],ap:"Beer foam
stabilisation, salad dressings, ice cream, fruit drinks",o:"Semi-synthetic (from seaweed
alginate)",w:"“Propylene glycol” in the name triggers consumer alarm out of proportion to the
chemistry.",s:"Permitted in EU/US with limits, chiefly in beer and dressings.",keep:null,r:[
{n:"Citrus fibre",d:"“Citrus fibre”",dose:"0.5–1.5% in dressings",m:"Partial",c:["Sauces &
dressings","Beverages"],g:"Food ingredient in EU/US.",t:"Covers dressings body; does not stabilise beer foam the
same way."},
{n:"Pectin (amidated / HM grades)",d:"“Pectin” (E440)",dose:"0.2–0.6% in acidified drinks",m:"Partial",c:
["Beverages","Sauces & dressings"],g:"EU: E440. US: GRAS.",t:"Good in acidic drinks; beer-foam use is not
equivalent — brewers usually accept PGA or drop the claim."}]},
/* ---- Colours ---- */
{e:"E102",n:"Tartrazine",f:["Colour"],ap:"Soft drinks, confectionery, desserts, sauces, snacks (lemon
yellow)",o:"Synthetic azo dye",w:"One of the “Southampton six”: in the EU, foods containing it must carry the
warning “may have an adverse effect on activity and attention in children”.",s:"EU: permitted with the mandatory
activity/attention warning. US: permitted as FD&C Yellow 5 with name declaration.",keep:null,r:[
{n:"Colouring food: turmeric / curcumin",d:"“Turmeric extract” or “colour: curcumin” (E100)",dose:"0.01–0.1% —
shade-guided",m:"Partial",c:["Beverages","Confectionery","Sauces & dressings"],g:"E100 authorised in EU;
turmeric is a standard US colour (exempt category).",t:"Less light-stable than tartrazine and can fade; slight
earthy note at high levels; pH-sensitive."},
{n:"Lutein (marigold extract)",d:"“Lutein” / “marigold extract” (E161b)",dose:"0.005–0.05%",m:"Partial",c:
["Beverages","Dairy & alternatives"],g:"EU: E161b with limits. US: limited colour uses — check
category.",t:"Oil-soluble; needs emulsification in drinks; cost is higher."}]},
{e:"E110",n:"Sunset yellow FCF",f:["Colour"],ap:"Orange drinks, sweets, jellies, snacks, sauces",o:"Synthetic
azo dye",w:"A Southampton-six colour: EU packs must carry the activity/attention warning.",s:"EU: permitted with
warning label. US: FD&C Yellow 6.",keep:null,r:[
{n:"Colouring foods: paprika + annatto blends",d:"“Paprika extract” / “annatto” (E160c / E160b) or colouring-
food concentrates",dose:"0.02–0.2% — shade-guided",m:"Partial",c:["Snacks","Sauces &
dressings","Confectionery"],g:"Both widely authorised (E160b/E160c; US exempt colours).",t:"Orange shade
achievable, but hue varies with pH and fat content; light stability is lower."},
{n:"Beta-carotene",d:"“Beta-carotene” (E160a) — a name consumers recognise from carrots",dose:"0.005–
0.05%",m:"Partial",c:["Beverages","Dairy & alternatives","Confectionery"],g:"EU: E160a. US: exempt-from-
certification colour, GRAS nutrient.",t:"Yellow-orange rather than true orange; needs emulsion systems in
drinks."}]},
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{e:"E129",n:"Allura red AC",f:["Colour"],ap:"Drinks, confectionery, desserts, cereals, sauces (bright red)",o:"Synthetic azo dye",w:"A Southampton-six colour — EU packs need the activity/attention warning, which alone drives most reformulation.",s:"EU: permitted with warning. US: FD&C Red 40, the most-used US red.",keep:null,r:[
{n:"Colouring food: black carrot concentrate",d:"“Black carrot concentrate” / “vegetable juice (for colour)” in the US",dose:"0.1–1% — shade-guided",m:"Partial",c:["Beverages","Confectionery"],g:"Colouring foods are ingredients (not additive colours) in the EU when not selectively extracted; US: vegetable juice colour is exempt.",t:"Pink-red, pH-dependent (best in acidic products); less heat-stable than Allura red; strawberry-type shades rather than pillar-box red."},
{n:"Beetroot red",d:"“Beetroot red” / “beet juice concentrate” (E162)",dose:"0.05–0.5%",m:"Partial",c:["Confectionery","Dairy & alternatives","Beverages"],g:"EU: E162. US: beet juice is an exempt colour.",t:"Fades with heat and light; earthy note if overdosed; poor in baked goods."},
{n:"Lycopene (tomato extract)",d:"“Lycopene” / “tomato extract” (E160d)",dose:"0.005–0.05%",m:"Partial",c:["Sauces & dressings","Beverages"],g:"EU: E160d with limits. US: permitted colour in specified foods.",t:"Orange-red, oil-soluble; better heat stability than beetroot but higher cost."}]},
{e:"E133",n:"Brilliant blue FCF",f:["Colour"],ap:"Drinks, confectionery, ice cream, desserts (bright blue; also blended for greens/purples)",o:"Synthetic",w:"A visibly artificial blue with no common natural counterpart — a frequent clean-label target in children's products.",s:"Permitted in EU (E133) and US (FD&C Blue 1).",keep:null,r:[
{n:"Spirulina extract (phycocyanin)",d:"“Spirulina extract” — declared as an ingredient/colouring food in the EU; “spirulina extract” colour in the US",dose:"0.05–0.5% — shade-guided",m:"Partial",c:["Beverages","Confectionery","Dairy & alternatives"],g:"US: spirulina extract is an approved exempt colour for many foods. EU: sold as a colouring food (not an additive colour).",t:"The standard natural blue, but heat- and acid-sensitive: best in cold, near-neutral products; fades in baking and hot-fill."},
{n:"Butterfly pea flower extract",d:"“Butterfly pea flower extract”",dose:"0.05–0.3%",m:"Partial",c:["Beverages"],g:"US: approved colour additive (2021) for many drinks/foods. EU: novel-food status must be checked market by market.",t:"Blue-to-purple pH showpiece in drinks; fades in acid over shelf life."}]},
{e:"E171",n:"Titanium dioxide",f:["Colour"],ap:"Confectionery coatings, chewing gum, cake decorations, sauces (brilliant white opacifier)",o:"Mineral (synthetic pigment grade)",w:"EFSA concluded in 2021 that E171 could no longer be considered safe as a food additive (genotoxicity could not be excluded), and brands worldwide began removing it.",s:"No longer authorised in EU foods since 2022 (removed from the Union list). Still permitted in the US (up to 1% by weight) and some other markets — a genuine EU/US divergence.",keep:null,r:[
{n:"Calcium carbonate",d:"“Calcium carbonate” (E170)",dose:"0.1–1% in coatings — opacity-guided",m:"Partial",c:["Confectionery","Bakery"],g:"EU: E170 authorised. US: GRAS.",t:"Less brilliant white than TiO2; higher doses can taste chalky and affect texture."},
{n:"Rice starch / native starch opacifier systems",d:"“Rice starch” (ingredient declaration)",dose:"1–5% in sauces/batters",m:"Partial",c:["Sauces & dressings","Bakery"],g:"Conventional food ingredient.",t:"Gives opacity in wet systems, not the hard-pan coating brilliance of TiO2."},
{n:"Starch-based white coating systems (supplier blends)",d:"Declared by components (starches, calcium salts)",dose:"Supplier-guided",m:"Partial",c:["Confectionery"],g:"Component-based; verify each component's status.",t:"Closest commercial route for sugar-coated products; finish is slightly less glossy."}]},
{e:"E150c",n:"Ammonia caramel (caramel colour class III)",f:["Colour"],ap:"Colas and dark drinks, sauces, gravies, baked goods, pet food",o:"Made by heating sugars with ammonia compounds",w:"“Ammonia caramel” reads badly on a label; its 4-MEI by-product has been a public concern (California Prop 65 listing for 4-MEI).",s:"Permitted in EU (E150c) and US; plain caramel (E150a) needs no ammonia/sulfite reactants.",keep:null,r:[
{n:"Plain caramel (class I)",d:"“Plain caramel” / “caramel colour” (E150a)",dose:"0.1–2% — shade-guided",m:"Partial",c:["Beverages","Sauces & dressings"],g:"EU: E150a. US: caramel colour is a standard exempt colour.",t:"Lighter colour strength — dark cola shades may need more, with flavour impact."},
{n:"Malt extract / burnt sugar (sugar colour by ingredient name)",d:"“Barley malt extract” or “burnt sugar syrup”",dose:"0.5–3% of the recipe, shade-guided",m:"Partial",c:["Bakery","Sauces & dressings"],g:"Conventional food ingredients; note barley = gluten allergen declaration for malt extract.",t:"Best in bakery/savoury systems; adds flavour and sugars."}]},
/* ---- Sweeteners ---- */
{e:"E950",n:"Acesulfame K (acesulfame potassium)",f:["Sweetener"],ap:"Zero-sugar drinks, yoghurts, desserts, chewing gum, tabletop sweeteners",o:"Synthetic",w:"A chemical-sounding name consumers increasingly avoid in “no artificial sweeteners” positioning.",s:"Permitted in EU (E950) and US with category limits; almost always blended with sucralose or stevia for taste.",keep:null,r:[
{n:"Steviol glycosides (stevia)",d:"“Steviol glycosides” / “stevia extract” (E960 — a natural-source sweetener that still carries an E-number)",dose:"0.01–0.05% for high-purity Reb A/M grades — sweetness-guided",m:"Partial",c:["Beverages","Dairy & alternatives","Confectionery"],g:"EU: E960 authorised. US: high-purity steviol glycosides are GRAS (whole-leaf/crude extracts are not).",t:"Licorice/bitter aftertaste in some grades (Reb M is cleaner); provides sweetness only — bulking must come from elsewhere."},
{n:"Monk fruit extract (luo han guo)",d:"“Monk fruit extract”",dose:"0.01–0.05%",m:"Partial",c:["Beverages","Dairy & alternatives"],g:"US: GRAS. EU: novel-food authorisation has been pending/restricted — verify before EU launch.",t:"Clean taste but expensive and supply-limited; usually blended with erythritol or stevia."}]},
{e:"E951",n:"Aspartame",f:["Sweetener"],ap:"Diet soft drinks, sugar-free gum, yoghurts, desserts, tabletop sweeteners",o:"Synthetic (dipeptide ester)",w:"The most debated sweetener: in 2023 IARC classed it “possibly carcinogenic” (Group 2B) while JECFA kept the ADI — the headlines, not a ban, drive reformulation. EU labels must also state “contains a source of phenylalanine”.",s:"Still permitted in EU (E951, with the phenylalanine statement) and US; not heat-stable, so absent from baked goods.",keep:null,r:[
{n:"Steviol glycosides (stevia)",d:"“Steviol glycosides” (E960)",dose:"0.01–0.05%",m:"Partial",c:["Beverages","Dairy & alternatives","Confectionery"],g:"EU: E960. US: GRAS (high-purity).",t:"Unlike aspartame it is heat-stable (usable in baking); taste profile differs — re-profile flavourings."},
{n:"Monk fruit extract",d:"“Monk fruit extract”",dose:"0.01–0.05%",m:"Partial",c:["Beverages"],g:"US: GRAS. EU: check novel-food status.",t:"Cost and availability are the constraints."},
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{n:"Erythritol blends (with stevia/monk fruit)",d:"“Erythritol” (E968) plus the high-intensity partner",dose:"Erythritol 0.5–5% to restore bulk in dry mixes/desserts",m:"Partial",c:["Bakery","Confectionery","Dairy & alternatives"],g:"EU: E968. US: GRAS.",t:"Erythritol alone is ~70% as sweet as sugar with a cooling effect; polyol laxation labelling may apply at high levels in the EU."}]},
{e:"E955",n:"Sucralose",f:["Sweetener"],ap:"Drinks, bakery, dairy, sauces, tabletop (heat-stable workhorse)",o:"Synthetic (chlorinated sucrose derivative)",w:"“Made from sugar” positioning has not protected it from artificial-sweetener avoidance; several brands market “no sucralose”.",s:"Permitted in EU (E955) and US; heat-stable, unlike aspartame.",keep:null,r:[
{n:"Steviol glycosides (stevia)",d:"“Steviol glycosides” (E960)",dose:"0.01–0.05%",m:"Partial",c:["Beverages","Bakery","Dairy & alternatives"],g:"EU: E960. US: GRAS (high-purity).",t:"Also heat-stable, so bakery uses transfer; onset of sweetness is slower — blends taste rounder."},
{n:"Monk fruit extract",d:"“Monk fruit extract”",dose:"0.01–0.05%",m:"Partial",c:["Beverages","Dairy & alternatives"],g:"US: GRAS. EU: verify novel-food status.",t:"Premium cost; often blended with stevia to balance taste and price."}]},
{e:"E954",n:"Saccharin",f:["Sweetener"],ap:"Tabletop sweeteners, drinks, canned fruit, low-cost sugar-free products",o:"Synthetic (one of the oldest artificial sweeteners)",w:"A legacy name with a dated safety debate in consumers' minds; rarely chosen for new clean-positioned products.",s:"Permitted in EU (E954) and US with limits; the historical US warning label was removed in 2000.",keep:null,r:[
{n:"Steviol glycosides (stevia)",d:"“Steviol glycosides” (E960)",dose:"0.01–0.05%",m:"Partial",c:["Beverages","Confectionery"],g:"EU: E960. US: GRAS (high-purity).",t:"Removes saccharin's metallic aftertaste; cost rises."},
{n:"Monk fruit extract",d:"“Monk fruit extract”",dose:"0.01–0.05%",m:"Partial",c:["Beverages"],g:"US: GRAS. EU: verify novel-food status.",t:"Cleaner label story; supply is the limit."}]},
/* ---- Flavour enhancer ---- */
{e:"E621",n:"MSG (monosodium glutamate)",f:["Flavour enhancer"],ap:"Soups, snacks, seasonings, ready meals, sauces, processed meats",o:"Fermentation-produced (industrial fermentation of sugars) — yet perceived as synthetic",w:"“No MSG” is one of the oldest clean-label claims, rooted in outdated anecdote rather than current safety assessments; brands remove the declaration, not the glutamate.",s:"Authorised in the EU (E621, with a 10 g/kg general limit) and US (GRAS, declared). Honest framing: the swaps below keep glutamate — from foods — while changing the declaration.",keep:null,r:[
{n:"Yeast extract",d:"“Yeast extract” — a kitchen-plausible ingredient name",dose:"0.2–1% of recipe",m:"Strong",c:["Sauces & dressings","Snacks","Meat & seafood"],g:"A food ingredient in EU/US (not an additive).",t:"Contains natural glutamates — that is precisely why it works. The swap changes the declaration, not the glutamate. Adds yeasty depth; not for delicate products."},
{n:"Tomato powder / concentrated tomato",d:"“Tomato powder”",dose:"0.5–3% of recipe",m:"Partial",c:["Sauces & dressings","Snacks"],g:"Conventional food ingredient.",t:"Glutamate-rich but brings colour and acidity; suits savoury, tomato-compatible products only."},
{n:"Mushroom powder (shiitake / porcini)",d:"“Mushroom powder”",dose:"0.3–1.5% of recipe",m:"Partial",c:["Sauces & dressings","Meat & seafood","Snacks"],g:"Conventional food ingredient.",t:"Guanylate-rich umami synergy; earthy flavour and cost limit dose."}]},
/* ---- Acidity regulators, phosphates, anti-caking, humectant ---- */
{e:"E450",n:"Phosphate salts — diphosphates & triphosphates (E450/E451)",f:["Acidity regulator","Raising agent","Stabiliser & thickener"],ap:"Processed cheese, cooked meats and seafood (water binding), bakery (baking-powder acid), potato products",o:"Mineral-derived salts",w:"“Phosphate-free” is a growing meat/dairy brief; high dietary phosphate intake is also a nutrition talking point.",s:"Permitted with category maxima in EU/US; total phosphate in meat products is capped. Functionally hard to replace in emulsified cheese and injected meats.",keep:null,r:[
{n:"Citrus fibre + sodium bicarbonate water-binding system",d:"“Citrus fibre” and “sodium bicarbonate” (E500 — a store-cupboard name)",dose:"Citrus fibre 0.5–2%; bicarbonate adjusted to pH target",m:"Partial",c:["Meat & seafood"],g:"Citrus fibre is a food ingredient; E500 is universally permitted.",t:"Recovers part of the water binding in comminuted meats; cook loss and bite still differ — expect yield trials, and processed cheese generally cannot go fully phosphate-free."},
{n:"Potassium citrate systems",d:"“Potassium citrate” (E332) in place of sodium phosphates in some cheese/dairy systems",dose:"0.5–2% in processed cheese trials",m:"Partial",c:["Dairy & alternatives"],g:"EU: E332. US: GRAS.",t:"Still an E-number salt — the win is “phosphate-free” positioning, not full clean label; melt behaviour needs rework."}]},
{e:"E338",n:"Phosphoric acid",f:["Acidity regulator"],ap:"Cola drinks (signature tang), processed cheese, jams",o:"Mineral acid",w:"Linked in consumer media to phosphate load and dental erosion debates in colas; some brands switch acidulants.",s:"Permitted in EU (E338) and US; defines cola flavour, so swaps change the product's signature.",keep:null,r:[
{n:"Citric acid",d:"“Citric acid” (E330) — the most recognised acid on any label",dose:"Acidify to the same target pH — typically 0.05–0.2% in drinks",m:"Partial",c:["Beverages"],g:"EU: E330, quantum satis in drinks. US: GRAS.",t:"Cleaner, fruitier acid profile — the drink will taste less like a classic cola; flavour house work needed."},
{n:"Lemon juice concentrate",d:"“Lemon juice concentrate”",dose:"0.2–1% to target acidity",m:"Partial",c:["Beverages","Sauces & dressings"],g:"Conventional food ingredient.",t:"Adds juice content claims potential, but acidity per ml is low and variable."}]},
{e:"E551",n:"Silicon dioxide (silica)",f:["Anti-caking & humectant"],ap:"Powders — spices, salt, icing sugar, drink mixes, supplements (free-flow agent)",o:"Mineral (synthetic amorphous silica)",w:"An unfamiliar mineral name on powder labels; “silica-free” appears in supplement and spice briefs.",s:"Permitted in EU (E551) and US (up to 2% in many foods). EFSA re-evaluation noted data gaps on nano-fractions — a watch item, not a ban.",keep:null,r:[
{n:"Rice-derived anti-caking (rice hull / rice flour systems)",d:"“Rice hull concentrate” or “rice flour” (ingredient declaration)",dose:"0.5–2% of the powder",m:"Partial",c:["Snacks","Bakery"],g:"Food-ingredient status in US blends; EU availability/naming varies — verify with supplier.",t:"Less effective than silica in humid supply chains and supply is limited; caking claims must be re-validated in storage tests."},
{n:"Calcium carbonate / tricalcium phosphate",d:"“Calcium carbonate” (E170) or “tricalcium
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phosphate" (E341)",dose:"0.5–2% of the powder",m:"Partial",c:["Bakery","Snacks"],g:"Both authorised (E170/
E341).",t:"Still E-numbers, but familiar mineral names; performance is humidity-dependent."}]},
{e:"E420",n:"Sorbitol",f:["Anti-caking & humectant","Sweetener"],ap:"Sugar-free gum and mints, toothpaste-
adjacent foods, diabetic products, humectant in baked goods",o:"Made by hydrogenating glucose (plant-derived
feedstock)",w:"Polyol awareness is rising ("sugar alcohols" on labels); digestive tolerance drives some
swaps.",s:"Permitted in EU (E420) and US (GRAS). EU requires a laxative-effect warning on foods with >10% added
polyols.",keep:null,r:[
{n:"Erythritol",d:""Erythritol" (E968)",dose:"Replace sorbitol on sweetness/bulk by trial — often 1:1 in dry
mixes",m:"Partial",c:["Confectionery","Bakery"],g:"EU: E968; the >10% polyol laxation statement also applies.
US: GRAS.",t:"Best-tolerated polyol (mostly absorbed), but strong cooling effect and crystallisation in some
systems; weaker humectancy than sorbitol."},
{n:"Xylitol",d:""Xylitol" (E967)",dose:"1:1 by weight in gum/mints trials",m:"Partial",c:
["Confectionery"],g:"EU: E967. US: GRAS for specified uses.",t:"Closest taste match (same sweetness as sugar)
with a dental-health story — but toxic to dogs (flag for pet-adjacent products) and costs more."},
{n:"Allulose (rare sugar) — US-led option",d:""Allulose"",dose:"1:1 bulk replacement in trials",m:"Partial",c:
["Bakery","Beverages"],g:"US: GRAS and exempt from "added sugars" declaration. EU: novel food — NOT yet
generally authorised; check status before EU use.",t:"Sugar-like behaviour with ~0.4 kcal/g; regulatory
asymmetry between US and EU is the main catch."}]},
/* ---- Low-concern "keep" examples ---- */
{e:"E330",n:"Citric acid",f:["Acidity regulator","Antioxidant"],ap:"Almost everything acidic — drinks, jams,
sauces, canned vegetables (acidulant and antioxidant synergist)",o:"Nature-identical — produced by fermentation
(Aspergillus niger on sugars); the same acid that makes lemons sour",w:"Occasionally swept into blanket "remove
all E-numbers" briefs — a good example of why blanket rules are bad formulation policy.",s:"Authorised without
numerical limit in most foods (EU quantum satis; US GRAS).",keep:"Keep as-is. Citric acid is fermentation-
produced, naturally present in citrus fruit, and recognised by shoppers when declared by name. Removing it gains
nothing on the label ("citric acid" IS the friendly declaration) and loses pH control and antioxidant synergy.
If a brief demands action, the only honest alternative is lemon juice concentrate — weaker, variable and
flavour-active.",r:[]},
{e:"E300",n:"Ascorbic acid (vitamin C)",f:["Antioxidant","Acidity regulator"],ap:"Fruit juices, cured-meat
colour support, flour treatment, canned fruit (anti-browning)",o:"Nature-identical — industrially synthesised/
fermented to be identical to vitamin C in food",w:"Rarely a removal target: it can be declared as "vitamin C" in
many contexts, which consumers read as a nutrient, not an additive.",s:"Authorised widely (EU E300; US GRAS).
Doubly useful: it is both an additive function and a vitamin.",keep:"Keep as-is. Ascorbic acid is vitamin C:
shoppers read it as a nutrient, regulators treat it as one of the lowest-concern additives, and it actively
protects colour and flavour. A "no E300" brief should be challenged — the swap would be to weaker, costlier
antioxidant systems for zero label gain.",r:[]}
];

/* ---------- Core logic ---------- */
function normE(s){ return String(s==null?"":s).toLowerCase().replace(/[\s\-_]/g,"").replace(/^e(?=\d)/,""); }
function findAdditive(e){ for(var i=0;i<DB.length;i++){ if(normE(DB[i].e)===normE(e)) return DB[i]; } return
null; }
function allSwapsCount(){ var t=0; for(var i=0;i<DB.length;i++) t+=DB[i].r.length; return t; }
function repText(rp){ return (rp.n+" "+rp.d+" "+rp.g+" "+rp.t).toLowerCase(); }
function searchDB(q){
  var needle = String(q==null?"":q).trim().toLowerCase();
  if(!needle) return DB.slice();
  var nE = normE(needle);
  return DB.filter(function(a){
    if(normE(a.e)===nE) return true;
    if(a.n.toLowerCase().indexOf(needle)>-1) return true;
    if(a.f.join(" ").toLowerCase().indexOf(needle)>-1) return true;
    for(var i=0;i<a.r.length;i++){ if(repText(a.r[i]).indexOf(needle)>-1) return true; }
    return false;
  });
}
function filterByFunction(list, fn){
  if(!fn || fn==="All") return list.slice();
  return list.filter(function(a){ return a.f.indexOf(fn)>-1; });
}
function repFit(rp, cat){
  if(!cat || cat==="All") return {fit:true, known:false};
  return {fit: rp.c.indexOf(cat)>-1, known:true};
}
function planKey(e, repName){ return normE(e)+"|"+repName; }
function planAdd(plan, e, repName){
  var a = findAdditive(e); if(!a) return {plan:plan, added:false, reason:"unknown additive"};
  var rp=null; for(var i=0;i<a.r.length;i++){ if(a.r[i].n===repName) rp=a.r[i]; }
  if(!rp) return {plan:plan, added:false, reason:"unknown replacement"};
  var k = planKey(e, repName);
  for(var j=0;j<plan.length;j++){ if(plan[j].key===k) return {plan:plan, added:false, reason:"duplicate"}; }
  var next = plan.slice();
  next.push({key:k, e:a.e, additive:a.n, rep:rp.n, decl:rp.d, dose:rp.dose, match:rp.m});
  return {plan:next, added:true, reason:"ok"};
}
function planRemove(plan, key){
  return plan.filter(function(row){ return row.key!==key; });
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}
function csvCell(v){ v=String(v==null?"":v); return /[",\n]/.test(v) ? '"'+v.replace(/"/g,'""')+'"' : v; }
function planToCSV(plan){
  var lines = ["Additive E-number,Additive,Replacement,Label declaration,Indicative dosage,Functional match"];
  for(var i=0;i<plan.length;i++){ var r=plan[i];

lines.push([csvCell(r.e),csvCell(r.additive),csvCell(r.rep),csvCell(r.decl),csvCell(r.dose),csvCell(r.match)].jo
in(",")); }
  return lines.join("\n");
}
function planToText(plan){
  if(!plan.length) return "Reformulation plan is empty.";
  var out = ["CLEAN-LABEL REFORMULATION PLAN ("+plan.length+" swap"+(plan.length===1?"":"s")+")"];
  for(var i=0;i<plan.length;i++){ var r=plan[i];
    out.push((i+1)+". "+r.e+" "+r.additive+"  ->  "+r.rep+" ["+r.match+" match]\n   Declare as: "+r.decl+"\n
Indicative dosage: "+r.dose); }
  out.push("Note: dosages are indicative starting points — validate with challenge tests and shelf-life studies;
confirm current regulations for your market.");
  return out.join("\n");
}
/*PURE-CORE-END*/
```